



FAIR 3D HERITAGE

*Documentation, Publication and Preservation
of the Digital 3D Heritage on the Web*

On the Scientific Validity of Virtual Reconstructions

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Abstract

Virtual reconstructions have become firmly established within architectural and urban research as instruments for both scholarship and public engagement. They visualise lost buildings, unrealised designs, earlier construction phases and even entire historical cities. Yet while these reconstructions are often highly persuasive in visual and medial terms, the underlying decision-making processes – the interpretation of sources, the weighing of alternatives and the assessment of plausibility – frequently remain insufficiently documented and difficult to access. This implicit knowledge, commonly referred to as paradata, is crucial for scholarly transparency, reusability and quality assurance.

IDOVIR (Infrastructure for Documentation of Virtual Reconstructions) addresses this long-standing deficit. Since 2023, IDOVIR has been available as a fully functional, openly accessible, web-based platform for documenting and communicating virtual reconstruction processes. It enables the structured recording of sources, arguments and modelling decisions, and makes these processes traceable and permanently accessible to both researchers and the wider public. In doing so, IDOVIR provides an essential tool for transparency, sustainability and methodological rigour in architectural research. The technical infrastructure, hosted by the University and State Library Darmstadt, ensures sustainable provision and long-term availability. To align IDOVIR with FAIR principles, a controlled vocabulary for reconstruction sources is being developed. A shared vocabulary for labelling sources (e.g. plans, photographs, paintings or invoices) is essential to make documentation of virtual reconstructions comparable and interoperable. Although some source categories are outlined in existing literature – such as architectural drawings, maps, photographs, material evidence, geographical data and textual descriptions – there

is currently no comprehensive classification specifically tailored to virtual reconstructions in architectural research.

Established vocabularies in art history, such as Iconclass or the Getty AAT, mainly address image databases and museum inventories, focusing on artistic technique or iconography. They do not adequately reflect the wide range of intermedial sources relevant to architectural reconstruction, including textual descriptions, data visualisations or archaeological evidence. Resources such as Wikidata, the GND or the Bamberg Vocabulary provide useful supplements, but none offer a holistic framework. Since reconstruction research draws on multiple disciplines, each with its own terminology, a comprehensive and integrative vocabulary is required.

The aim of the developed hierarchical classification scheme for reconstruction sources is to establish a standardised vocabulary that enables the systematic description and evaluation of sources with regard to their evidential value and reliability, supports comparative assessments of plausibility, facilitates automated (e.g., AI-based) categorisation. The classification is based on objectively defined main categories to ensure consistent and reproducible results. By labelling the sources with distinct identifiers for source types, comparability across different projects will be improved.

As the second cornerstone towards a FAIR-compliant platform, a mapping of the data to CIDOC CRM and HDTO (Heritage Digital Twin Ontology) is currently developed. By making the data available as Linked Open Data, it will be integrated into national and European infrastructures (e.g., NFDI, ECCCH) or can be queried directly via a SPARQL interface, and thus contributes directly to data reusability and sustainability.

By foregrounding paradata and embedding them within a robust digital infrastructure, IDOVIR contributes to a more transparent, reflexive and FAIR-oriented practice of 3D heritage research. The project demonstrates how virtual reconstructions can evolve from visually compelling representations into fully documented, critically assessable scholarly arguments.